

Halving goes to the ultraviolet; confinement lives in the infrared

Didactic note, 2026-09-28 (Claude, prompted by the user's question; §7, §3b and several refinements come from independent answers to the same questions by GPT-6 Astra and Claude Fable). Confinement and the mass gap are infrared properties, and distinct ones: a spectral bound for the gap, an area law for confinement. Halving the lattice moves toward the ultraviolet. The refinement programme of the atlas nevertheless bears on the gap, for five reasons, each stated with its source:

1. **One map, two directions.** A halving adds modes at the ultraviolet end; integrating them out again is one Wilson renormalization step toward the infrared. Refinement and coarse-graining are the same pushforward, Proposition 1 of the series/parallel note, read in opposite directions.
2. **The infrared is the invariant of a refinement.** A continuum limit holds the physical scales fixed. In lattice units every infrared quantity shrinks at each halving: am and $a\Lambda$ halve, $a^2\sigma$ quarters. The question a halving asks about the infrared is whether these stay constant in physical units.
3. **The exponential ladder selects what survives.** Along the one-loop trajectory a non-perturbative term $e^{-\gamma/t}$, with $t = \hbar g^2(a)$ (the atlas writes c for γ), equals $(a\Lambda)^{2b_0\gamma}$, so a halving multiplies it by $2^{-2b_0\gamma}$. A gap, $am \propto e^{-1/(2b_0\hbar g^2)}$, would be the term with $2b_0\gamma = 1$: it halves exactly when the lattice halves, and so survives in physical units. Every term with $2b_0\gamma > 1$ dies. The ladder fixes the size a surviving gap must have; that the coefficient C in $E = C\hbar c\Lambda$ is positive needs non-perturbative input (Astra).
4. **A growing ultraviolet end and a fixed infrared end.** From spacing a to the correlation length ξ there are $\log_2(\xi/a)$ halvings. Refining adds steps only at the ultraviolet end, where t is small; the steps where t is of order one, around $\beta_W \simeq 6$ for $SU(3)$ with $\xi/a \simeq 1.3$, are the same finite set for every a . Confinement is decided there once; refinement requires the growing number of ultraviolet steps to be controlled uniformly. That is one of two uniformities (Fable): in a , which the halvings address, and in the box L at fixed a , which is a mixing statement.
5. **Refinement exposes lattice artifacts.** Compact $U(1)$ has a lattice gap that refinement at fixed coupling removes. A property that survives every halving is a property of the continuum theory.

1. One map, two directions

Proposition 1 of the series/parallel note writes the law of the coarse links as the pushforward of the fine heat-kernel measure under the product of half-links. Read from coarse to fine it is a refinement: the fine measure is a consistent extension of the coarse one, with new modes at the shortest scale. Read from fine to coarse it is a blocking step: the mid-plane integral Ψ is the effective action generated by the modes of wavelength between a and $2a$. The same factor Ψ , and the same parallel defect, carry the renormalization in both readings. What the atlas calls “what survives refinement” is, in the other reading, “what is left of the ultraviolet after coarse-graining down to the scale of interest”.

2. The infrared is what a refinement holds fixed

A lattice at spacing a has three scales: a , the correlation length $\xi = \hbar/(mc)$ and the box L . A continuum limit sends $a/\xi \rightarrow 0$ at fixed ξ/L (or with $\xi/L \rightarrow 0$ afterwards). In $1 + 3$, one-loop running ties the bare coupling to the spacing through $a\Lambda = e^{-1/(2b_0t)}$, $t = \hbar g^2(a)$, $b_0 = 11N/(48\pi^2)$, so that Λ does not move while a does. The infrared quantities are ratios to Λ : m/Λ , σ/Λ^2 . A halving never computes them; it tests whether they remain what they were. This is the precise sense in which the programme's gap statement (AGENTS.md: which operator,

which limits in which order, what the gap is measured in) is a statement about all halvings at once.

In 1+2 the relation is simpler: the heat time $t = \lambda_3 a$ itself halves at every step, the ultraviolet steps are close to Gaussian, and the infrared scale is $\hbar c \lambda_3$ (the cell-7 target $E = C_3 \hbar c \lambda_3$ of the atlas), fixed by the dimensionful coupling.

3. The exponential ladder

At one loop, $e^{-\gamma/t} = (a\Lambda)^{2b_0\gamma}$, and $2b_0\gamma$ is the scaling dimension of the term (atlas §4, “The exponential ladder”, where γ is written c). A mass has dimension one, so a physical gap is the term that halves with the lattice: $am \propto (a\Lambda)^1$, the member of the ladder with $2b_0\gamma = 1$. The corrections a single step leaves sit far higher: the bridge image terms at $2b_0\gamma = 11N/3$ (roots) and $11N$ (the $SU(3)$ centre images of a trisection, centre note), beyond the per-volume threshold $2b_0\gamma = 4$ of the zero-spacing note. So every ultraviolet step contains the infrared scale as a tiny non-perturbative term of exactly the size that survives, and nothing else of its kind survives with it. This is dimensional transmutation read step by step. The term is invisible to perturbation theory, since $e^{-1/(2b_0\hbar g^2)}$ has a vanishing Taylor series in g^2 ; it is visible to the exact step, which is why the atlas insists on exact halvings. Suppressing the image terms is ultraviolet control; the gap and the area law are further, long-distance obligations (Astra).

3b. Dimensional transmutation: a scale, then a gap

The user’s follow-up question, answered by merging three independent answers (Claude, GPT-6 Astra and Claude Fable, 2026-09-28).

1. **The trade.** Classical Yang–Mills in 1 + 3 has one dimensionless coupling and no dimensionful constant, so every classical energy floor is 0 or ∞ (G07, Theorem 1). Quantization supplies $\hbar$ and the lattice supplies a ; the dimensionless $t = \hbar g^2$ builds no mass, so a mass must read $m = (\hbar/ac) f(t)$, and keeping it fixed as $a \rightarrow 0$ forces t to run: $\mu dt/d\mu = -2b_0 t^2 + O(t^3)$, whose integration constant is $\Lambda = \mu e^{-1/(2b_0 t(\mu))}$. One measured coupling fixes it; a change of scheme rescales Λ by a finite factor (zero-spacing note) and the pure numbers C_i inversely, so every physical energy $E_i = C_i \hbar c \Lambda$ is unchanged.
2. **Invisible to every order.** In lattice units $\delta(a) = aE/(\hbar c) = C e^{-1/(2b_0 t)}$, whose Taylor coefficients in t all vanish. Perturbation theory starts from massless gluons and keeps them massless; its series is asymptotic, with ambiguities of the size of powers of Λ (renormalons). With $\hbar$ in the exponent, $\Lambda \rightarrow 0$ faster than any power as $\hbar \rightarrow 0$ at fixed g_{cl} : the classical limit removes the gap.
3. **The ladder degenerates into the exponential.** Below four dimensions $\lambda_D = \hbar g_{cl}^2$ has dimension length $^{D-4}$, and the only energy is $\hbar c \lambda_D^{1/(4-D)} \propto \hbar^{(5-D)/(4-D)}$: $\hbar^{3/2}$ in 1 + 1, $\hbar^2$ in 1 + 2 (and $\hbar^2$ for the rotor, $\hbar^{4/3}$ for the quartic matrix model; the dimension ladder). There $t_n = \lambda_D a_n^{4-D}$ falls geometrically and the shifts sum to a finite renormalization. At $D = 4$ the exponent diverges, a^{4-D} becomes a logarithm, and the power of $\hbar$ becomes $e^{-1/(2b_0 \hbar g^2)}$, as $(g^2)^{1/\varepsilon}$ with $\varepsilon = 4 - D$ becomes an exponential in dimensional regularization. The renormalization group, beyond dimensional analysis, supplies this step.
4. **Supplied floor, generated gap.** $\hbar$ is supplied, written into $e^{-S/\hbar}$ and kept by every limit, including those that lose the gap (Gross’s photon; free Maxwell). The gap is generated, from two scale-free structures that fail to commute (G07 §1; G08). Below four dimensions the classical coupling carries the unit; in 1 + 3 transmutation manufactures it from $\hbar$ non-algebraically, which is why a 1 + 3 gap together with c returns no action unit (G08’s action-critical case), unlike G , c and l_P .
5. **A scale is not yet a gap.** The gap needs $\inf C_i > 0$ over the states above the vacuum that gauge-invariant observables see, uniformly in L and a , on a constructed continuum

measure. Theories with a scale and no gap show the difference: massless QCD has Λ and, with spontaneous chiral symmetry breaking, massless pions (the joint paper’s benchmark); an asymptotically free theory flowing to an infrared fixed point has a Λ and a scale-invariant infrared. In mechanics, the two-dimensional delta potential transmutes $m\lambda/\hbar^2$ into a binding energy E_B with the free continuum $[0, \infty)$ above it; a field-theoretic gap needs in addition that the vacuum be the ground state and every excitation be massive: “transmutation supplies the scale and confinement supplies the gap” (G07, Proposition 12 and §6).

4. A growing ultraviolet end and a fixed infrared end

The trajectory from spacing a to the scale ξ has $\log_2(\xi/a)$ halvings. As $a \rightarrow 0$, every new halving is added at the ultraviolet end, where $t = \hbar g^2(a) \rightarrow 0$ and the atlas cells 1–6 give one-step control. The infrared end is the handful of order-one steps where t is of order one; for $SU(3)$ with the Wilson action these sit around $\beta_W \simeq 6$, a few steps above the correlation length (the confinement-scale bands note, with published numbers). That end is the same finite set for every a . Its analysis is the finite verification of the intermediate region; the uniform control of an unbounded number of ultraviolet steps feeding it is what Jaffe and Witten name as the missing idea. The division of labour in the atlas follows: cells 1–6 are the ultraviolet steps, cell 7 is the infrared end.

5. Where confinement can be lost

The compact $U(1)$ theory in $1+2$ is the proved example. Its lattice gap comes from monopoles; at fixed coupling the refinement removes it and the continuum limit is the free photon (Gross), while along the Göpfert–Mack trajectory the gap survives only with a diverging monopole density per physical volume (atlas §4; the monopole note). The ultraviolet step is where a lattice gap is exposed as a lattice artifact. In $1+3$ the same monopole bound, of order $(L/a)^4 e^{-\pi^2/(8\hbar g^2)}$, is useless, since $t = \hbar g^2$ stays fixed: with the coupling standing still, the ultraviolet direction simplifies nothing (Fable). For $SU(3)$ it helps only through asymptotic freedom: the coupling runs to zero logarithmically (cell 4), and the gap is expected to survive because it is tied to Λ , which the running holds fixed.

6. Newton’s parallel

In Newton’s cell the sagitta and the Galileo action vanish under refinement, $K_\tau \propto \tau^3$ (cut-measure note), while the force survives as the ratio those vanishing quantities define. On the lattice $am \rightarrow 0$ while m/Λ survives. In both cases refinement never computes the surviving quantity directly: it certifies that a ratio stays fixed while every per-cell quantity goes to zero. What refinement cannot remove in either case is the supplied action floor $\hbar$ (atlas §4, “The action floor survives what removes the mass gap”).

7. What the explanation leaves open

The two independent answers agree on what the paper must state plainly.

- **The ends have yet to meet** (Fable). The halvings can deliver, at best, a running coupling from small t down to $t \simeq \frac{1}{2}$. The steps across the meeting region lie outside a small-field expansion, and the intermediate band needs a certified mixing box of a few hundred to a few thousand links (confinement-scale bands), which no method provides today. The gap is proved at the coarse end and would be transported; the transport is the open part.

- **The one-step objects are uncontrolled for $SU(N)$** (both). Hypothesis $P(\alpha)$, its stability under iteration, uniform remainders and the matching term $c_{\text{in}} - c_{\text{out}}$ are open; the ladder exponents are formal.
- **The fine measure is a construction** (Astra). The pushforward preserves every coarse observable exactly, large Wilson loops included, $\langle F \rangle_{p*\mu} = \langle F \circ p \rangle_\mu$; a coarse measure alone leaves the fine conditional laws undetermined, so the refinement family must be built.
- **Coarse observables only** (Fable). Transport covers what coarse loops see. A gap of the fine transfer matrix over all local operators needs the lowest states to be visible to coarse loops, or a separate argument per channel.
- **Orders of limits and the rest of the problem** (both). The size clause ties L/a to t ; the paper must say which order it takes (for example the continuum first at fixed L , then $L \rightarrow \infty$). A complete argument also needs a nontrivial continuum theory, reconstruction, volume-uniform spectral bounds and, for confinement, its own area-law argument.

8. Consequence for STATE

None; this note explains the programme's logic for the joint paper and for readers of the atlas.